

UN1208

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## 1 Lecture 3/28/24

### 1.1 Tangent Plane

We have the following idea:

$$\mathbf{v} = T_p x$$

where  $T_p x$  is the set of all vectors tangent to curves in  $x$  passing through  $\mathbf{p}$

We can take the velocity to be the tangent line at the point. For different parameterizations, we can change the velocity.

If  $\mathbf{x}$  is  $k$  dimensional, then we have that  $T_p x \approx \mathbb{R}^k$

### 1.2 Tangent Bundle

$\mathbb{R}^n$ , let  $U \subset \mathbb{R}^n$ ,  $U$  open, contractible.  $U$  has a "natural" structure of a  $n$ -manifold.  $\forall \mathbf{p} \in U$  we have  $T_p U$  s.t.  $\mathbf{p} \implies T_p U \approx \mathbb{R}^n$ . We have that  $TU = U \times \mathbb{R}^n$  where  $TU$  is the tangent bundle of  $U$ , which has a strength of a  $2n$  dimensional manifold.

Now let  $X$  be a  $k$  manifold  $\in \mathbb{R}^n$ , and  $U$  is an open set containing  $X$  contractible in  $\mathbb{R}^n$ . We will define the tangent bundle to be  $TX \subset TU$  where

$$TX = \{(\mathbf{p}, \mathbf{v}) | \mathbf{p} \in X, \mathbf{v} \in T_p X\}$$

where  $TX$  is the tangent bundle in  $X$  with  $2k$  dimensional manifold.

Thus we have for a function  $F : X \rightarrow Y$  we have that  $DF : TX \rightarrow TY$ .

### 1.3 Vector Fields

Vector fields are maps  $F : R^n \rightarrow R^m$  where we take  $x \rightarrow F(x)$ . The use of the vector field is such that the dimensions maps vectors to vectors and there exists a vector at each point in the field. And we can see that vector fields can represent a section of the tangent bundle.

If  $F : U \rightarrow R^n$  is a vector field. It can be viewed as a map  $F : U \rightarrow TU$ . We can represent TU as a summation of vectors that are perpendicular and create a set of basis vectors  $(p, v_1) + (p, v_2) = (p, v_1 + v_2)$ . We can extend this idea to vector fields in  $X^k \subset U^n$

**Unit Circle** We have the circle  $S^1 \subset R^2$  where we define a vector field  $F : R^n \rightarrow R^n$  that takes  $(x_1, x_2) \rightarrow (-x_2, x_1)$  which creates a tangent bundle for the surface  $S^1$ . Thus this creates a vector field on  $S^1$

**Unit Sphere** Let's define the vector field  $F : R^3 \rightarrow R^3$  s.t. it maps from  $(x_1, x_2, x_3) \rightarrow (-x_2, x_1, 0)$  which creates a series of arrows that are perpendicular to the surface around the  $z$  axis. Where the norm of the vectors are going to be larger around the equator of the sphere while it approaches 0 as it reaches the poles of the  $z$  axis.

### 1.4 Lagrange Multipliers

Let us consider the drawing between a curve and a hollow cylinder where there exists an intersection between the two. Suppose the equation of the cylinder is  $g(x, y) = 1, g(x, y) = x^2 + y^2$  and the graph of the curve is going to be  $f : R^2 \rightarrow R$ . Our goal is to find the max/min of the values of  $(x, y)$  which are restricted or constrained to the intersection of  $g(x, y) = 1$

To approach this question, let us approach it via level curves in which the level curves exist in  $R^2$ . When you take the level curve when the function attains the maximum, there exists a point in which there exists exactly one point where this touches s.t. the level curve exactly touches  $g(x, y) = 1$ . We will denote this point as  $p = (x_0, y_0)$  where  $f$  achieves its max/min in the intersection. In this condition,

$$\nabla f = \lambda \nabla g$$

for same scalar  $\lambda$ .

**Theorem** Let  $X$  be a  $k$ -dimensional manifold in  $R^n$  obtained as the zero-level set  $G^{-1}(0)$ , where  $G : R^n \rightarrow R^{n-k}$  where  $k$  is the codimension where  $G = (g_1, g_2, \dots, g_{n-k})$ . For  $p \in X$  we wish for  $DG_p$  to be surjective in order for  $p$  to be regular

Let's say we have a function  $f : U \rightarrow R$  where  $U$  is an open set in  $R^n$ ,  $X \subset U$ . Suppose  $p$  is a point of max/min of  $f$  when restricted to  $X$ . Then there exists scalars  $\lambda_1, \lambda_2, \dots, \lambda_{n-k}$  s.t.  $\nabla f(p) = \nabla \lambda_1 g_1(p) \dots \nabla \lambda_{n-k} g_{n-k}(p)$

**Proof** If the map  $DG_p$  is surjective, then we have that

$$T_p X = \text{span}\{\nabla g_1(p)_1, \dots, \nabla g_{n-k}(p)_{n-k}\}^\perp$$

Lets say we have some  $v \in T_p X$ . There exists  $r : I \rightarrow X, 0 \in I$  open interval. Let us have  $r(0) = p, r$  is smooth and  $r'(0) = v$ .

Let us take the condition that  $p$  is a maximum of  $f(x) = f(p), \forall x \in X$ .  $0$  is a point of max for the function  $f \cdot r : I \rightarrow R$ .

$$\frac{d}{dt}(f \cdot r), r = 0 = 0 \implies \nabla f(r(0)) \cdot r'(0) = \nabla f(p) \cdot v$$

Thus we have that

$$\nabla f(p) \cdot v \in T_p X^\perp = \text{span}(\nabla g_1(p)_1, \dots, \nabla g_{n-k}(p)_{n-k})^\perp$$

## 2 Lecture 4/2/2024

### 2.1 Integration

We have  $[a, b], a < b, f$  is bounded on  $[a, b]$ .

In Calculus I, we assumed for  $f$  to be greater than one and develop an intuition through the area. However, here, we are going to develop a partition of  $[a, b]$  denoted as  $a = t_0 < t_1 \dots < t_n = b$

$$m_i = \inf f, i = 1, \dots, n$$

which means that  $m_i \leq f(x), \forall x \in t_0 \dots t_n$

Associated to the function  $f$  and the Partition  $P$ , we will denote

$$L(f, P) = \sum_{i=1}^n m_i(t_i - t_{i-1})$$

$$U(f, P) = \sum_{i=1}^n M_i(t_i - t_{i-1})$$

where  $L(f, P) \leq \sum_{i=1}^n f(x_i^*)(t_i - t_{i-1}) \leq U(f, P)$

We can say that  $Q$  is a refinement of the partition  $P$  if it has all the  $t_i$  in  $P$  and some more points ' ,

**Lemma** If  $Q \prec P$ , then  $L(f, P) \leq L(f, Q) \leq U(f, Q) \leq U(f, P)$  where the size of a subdivision  $P$  is  $\max(t_i - t_{i-1})$

**Lemma** If  $P$  and  $Q$  are subdivisions of  $[a, b]$ , then they have a common refinement  $R, R \prec Q, R \prec P$ .

**Definition** :  $L(f) = \sup\{L(f, P) | P \text{ partition of } [a, b]\}$  and define  $U(f)$  as the infimum.

**Definition**  $f$  bounded on  $[a, b]$  is Riemann integrable if  $L(f) = U(f)$ . We will write this as

$$\int_{[a,b]} = L(f) = U(f)$$

Thus we have a class of functions where bounded functions are integrable functions but not vice-versa. Now let us ask which type of functions are integrable.

**Lemma** Let  $f$  be a bounded and integrable function on the interval  $[a, b]$ . Then, for every  $\epsilon > 0$ , there exists a partition  $P$  of  $[a, b]$  such that  $U(f, P) - L(f, P) < \epsilon$ .

**Proof** Let  $L(f) = \sup\{L(f, P) | P \text{ is a partition of } [a, b]\}$ . Given  $\epsilon > 0$ , there exists a partition  $Q$  of  $[a, b]$  such that  $L(f) - L(f, Q) \leq \frac{\epsilon}{2}$ . Similarly, define  $U(f) = \inf\{U(f, P) | P \text{ is a partition of } [a, b]\}$ . The proof continues and is show in the notes.

**Theorem** If  $f$  is continuous on  $[a,b]$  then  $f$  is integrable on  $[a,b]$ .

**Proof**  $f$  continuous on  $[a,b] \implies f$  is uniformly continuous.  $\forall \epsilon > 0, \exists \delta > 0$  s.t. if  $|x - y| < \delta$  then  $|f(x) - f(y)| < \epsilon$ . From this we have

$$\sum_1^n m_i(t_i - t_{i-1}) - \sum_1^n M_i(t_i - t_{i-1})$$

From here, choose  $\epsilon > 0$ . Let  $P$  be partition of  $[a,b]$  such that the size of  $P$  is  $< \delta$ . Then when we look at  $U - L$ , we get that this is going to be equal to

$$\sum f((x_i^*) - f(y_i^*))(t_i - t_{i-1})$$

where this is going to be less than  $\sum \epsilon \delta$ .

What we show here is that all continuous functions on  $[a,b]$  are also integrable and thus exist inside integrable functions.

**Lemma** Let  $f$  be bounded on  $[a,b]$  and let  $c \in (a,b)$ . Suppose that for any  $\delta > 0$  so that  $(c - \delta, c + \delta) \subset [a,b]$  and  $f$  is integrable on  $[a, c - \delta]$  and  $[c + \delta, b]$ , then  $f$  is integrable on  $[a, b]$

**Proof** We must first show that the partition of  $c$  must be integrable. Thus we can do this by simply creating the partition of  $c$ .

**Corollary** For piecewise functions on  $[a,b]$ , the function is also integrable since it is continuous and bounded on  $[a,b]$  except for finitely many points in  $[a,b]$

## 2.2 Fundamental Theorem of Calculus

If  $f$  is continuous on  $[a,b]$  and is therefore integrable, then the function

$$F(x) = \int_{a,x} f$$

is continuous on  $[a,b]$ , differentiable on  $(a,b)$  and  $F'(x) = f(x)$

**Proof** We need to first show that  $F$  is continuous on  $[a,b]$ . Given that  $f$  is continuous on  $[a,b]$  implies that  $|f|$  is continuous on  $[a,b]$ . Let  $c > 0$  s.t.  $|f(x)| < c, \forall x \in [a, b]$  Thus, let  $\epsilon > 0, \delta = \epsilon/c$ . We will get that

$$|F(x) - F(y)| = \left| \int_{a,x} f - \int_{a,y} f \right|$$

WLOG, let  $x \leq y$ . We get that

$$\left| \int_{x,y} f \right| \leq \int_{x,y} |f| \leq \int_{x,y} c \leq c\delta = \epsilon$$

### 3 Lecture 4/9/24

#### 3.1 Multiple Integration

##### 3.1.1 Fubini's Theorem [practical version]

Let us have a map  $f : R^n \rightarrow R$  which is continuous and integrable. The domain will be denoted as  $\mathbf{x}, \mathbf{y} \in R^p, R^q; p + q = n$ . For any  $x \in R^p$ ,

$$x \rightarrow \int_{R^q} f(x, y) dy$$

is well defined and so is the respective opposite variables.

Both functions are integrable and the integral over  $R^n$

$$\int_{R^n} f(x, y) = \int_{R^p} \int_{R^q} f(x, y) dy dx$$

This practical version allows us to switch the integrals such that we now have  $dx dy$ .

**Remark** For  $f(x, y)$  continuous on  $R^p \times R^q$  compactly supported implies that both  $x$  and  $y$  are continuous and compactly supported on their respective regions.

### 3.1.2 Fubini's Theorem [generalization]

For a function  $f$  bounded and integrable over  $R^n$ , we will have that for the same given conditions: For any  $x \in R^p$ ,

$$x \rightarrow \int_{R^q} f(x, y) dy$$

$$x \rightarrow \int_{R^q}^- f(x, y) dy$$

are Riemann integrable on  $R^p$  and the following holds

$$\int_{R^n} f(x, y) = \int_{R^p} \int_{R^q} f(x, y) dy dx = \int_{R^p} \int_{R^q}^- f(x, y) dy dx$$

For  $A = [a, b] \times [c, d]$ ,  $f : A \rightarrow$

### 3.1.3 Elementary Regions in $R^2$

SO lets say we have a type I region. We will have that for  $D = (x, y) \in R^n$ ,  $a \leq x \leq b$  we will have that  $g(x) \leq y \leq f(x)$  where both functions are continuous over the interval.

$$\int \int_D f(x, y) dx dy = \int_a^b \int_{g(x)}^{h(x)} f(x, y) dy dx$$

Type II functions are going to be x-simple regions where the graphs will not cross but are running along the y direction.

$$\int \int_D f(x, y) dx dy = \int_c^d \int_{g(y)}^{h(y)} f(x, y) dx dy$$

### 3.1.4 Elementary Regions in $R^3$

For regions in  $R^3$  that are  $z$  simple are going to be defined as the following:

$$\int \int \int_D f(x, y, z) dx dy dz = \int \int_D \left( \int_{g(x, y)}^{h(x, y)} f(x, y, z) dz \right) dx dy$$

And if our function is a measurable set, then our integral is also a functional set where some discontinuities are allowed.

### 3.2 The Change in Variable Theorem

Recall that for a function  $f : R \rightarrow R$  where we will have two maps that take the function from a real line to a real line. Defined these lines to be  $\phi \rightarrow \phi(a) - \phi(b)$  which takes the function from (a,b) to (r,s)

Thus if  $F = f$ , then  $\cdot\phi(u) = (f \cdot \phi)(u)$ . The idea is that we have found a  $C_1$  map that has take us from the function.

Let  $f : K \rightarrow R, k$  compact in  $R^n$  with  $f$  continuous on  $k$ .

Let  $K' \subset R^n$  be a compact set and  $u$  be an open nbhd of  $k' \in R^n$ . Now that we have our compact, we define  $f$  to be a a continuous function taking  $k \in R^n$  to  $R$ . Now we also have the function  $F$  that takes  $k' \in U$  and takes it into  $k$  such that  $F(k') = K$ . We would also like that  $DF(\mathbf{x})$  for any  $x \in k$  to be injective. From this we have the following relationship:

$$\int_k f(x)dx = \int_{k'} (f \cdot F)(y)|DF(y)|dy$$

where  $|DF(y)|$  is the determinant of the Jacobian of  $F$ .

**Change of Variable Theorem** Recall the 1-dimension equivalent. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  and let  $\varphi$  be a  $C^1$  map. Let  $f$  be continuous on  $[r, s]$ . We wish to find  $\int_r^s f(x)dx$ . Consider  $x = \varphi(u)$ . Let  $\varphi(\{a, b\}) = \{r, s\}$  with  $a < b$  and  $r < s$ . Then,  $f \circ \varphi : [a, b] \rightarrow \mathbb{R}$ . If  $F' = f$ , then  $(F \circ \varphi)'(u) = (F' \circ \varphi)(u)\varphi'(u) = (f \circ \varphi)(u)\varphi'(u)$ . Then,

$$\int_r^s f(x)dx = \int_a^b (f \circ \varphi)(u) |\varphi'(u)| du$$

For multiple dimensions, let  $f : K \rightarrow \mathbb{R}$ , where  $K$  is compact on  $\mathbb{R}^n$  and  $f$  is continuous on  $K$ . Let  $K' \subset \mathbb{R}^n$  be a compact set and  $U$  be an open neighborhood of  $K'$  in  $\mathbb{R}^n$ . Let  $F : U \rightarrow \mathbb{R}^n$  be a  $C^1$  map such that  $F(K') = K$ . Let  $DF(\vec{y})$  be injective for any  $\vec{y} \in K'$ . Then,



$$\int_K f(\vec{x}) d\vec{x} = \int_{K'} (f \circ F)(\vec{y}) |\det DF(\vec{y})| d\vec{y}$$

where  $|\det DF(\vec{y})|$  is absolute value of the determinant of the Jacobian of  $F$  at  $\vec{y}$ .

## 4 Lecture 4/11/2024

Consider  $X^k \subset \mathbb{R}^n$ , where  $X$  is a  $k$ -dimensional (smooth) manifold in  $\mathbb{R}^n$ . We will work with properties of manifolds.

**Definition** For a circle in  $\mathbb{R}^2$  of radius  $r$ , the curvature at a point on the circle is defined to be  $\kappa = \frac{1}{r}$ .

**Parametrizations** For  $k = 1$ , one-dimensional manifolds are curves. This is  $\vec{r} : [a, b] \rightarrow \mathbb{R}^n$ , where  $t \mapsto \vec{r}(t) = (x_1(t), \dots, x_n(t))$ .  $\vec{r}$  is referred to as the position vector. We can also consider  $\frac{d\vec{r}}{dt} = \vec{t}(t)$  (if  $\vec{r}$  is of class  $C^1$ ), which is the velocity vector, and also  $\frac{d\vec{v}}{dt} = \vec{a}(t)$  (if  $\vec{r}$  is of class  $C^2$ ), which is the acceleration vector.  $C$ , the image of  $\vec{r}$ , is a space curve, with initial point  $\vec{p}$  and final point  $\vec{q}$ .  $C$  is a simple curve if  $\vec{r}$  is injective on  $(a, b)$ .  $C$  is a closed curve if  $\vec{p} = \vec{q}$ .  $C$  is a simple closed curve if  $C$  is both simple and closed.  $\vec{p}$  and  $\vec{q}$  are boundary points. Denote  $\partial C = \{-\vec{p}, +\vec{q}\}$ , where the negative denotes the start and the positive denotes the end. Note that for a curve,  $\vec{v}$  is not an intrinsic property. We can however define  $\vec{T}(t)$  to be the unit tangent to  $C$ . Up to sign,  $\vec{T}(t)$  is intrinsic to the curve.

**Example** Consider  $\vec{r} : [-1, 1] \rightarrow \mathbb{R}^2$ , where  $\vec{r}(t) = (t, t^2)$ . This is the parabola  $y = x^2$  restricted to  $-1 \leq x \leq 1$ , with the orientation from left to right. Note that  $\vec{r} : [-1, 1] \rightarrow \mathbb{R}^2$ , where  $\vec{r}(t) = (t^2, t^4)$  only has the right half of the parabola. The resulting curve is no longer simple.

For reparametrizations, we are interested in maps  $\varphi : [c, d] \rightarrow [a, b]$ , where  $\varphi$  is one-to-one. If  $\varphi' > 0$  then  $\varphi$  is orientation preserving. If  $\varphi' < 0$  then  $\varphi$  is orientation reversing.

For  $k > 1$ , then we have  $X^k$ , which is a  $k$ -dimensional manifold in  $\mathbb{R}^n$ , where  $\partial X \neq \emptyset$ . (The manifold has a boundary).

**Definition**  $X$  is a  $k$ -dimensional manifold with a boundary in  $\mathbb{R}^n$  if for any  $\vec{p}$  in  $X$ , either

1. there exists an open set  $U \subseteq \mathbb{R}^k$  and  $H : U \rightarrow X$  is a one-to-one  $C^1$  map satisfying  $H(U) \ni \vec{p}$  (in other words, if a neighborhood of  $\vec{p}$  is homeomorphic to  $\mathbb{R}^k$ ),
2. or a neighborhood of  $\vec{p}$  in  $X$  is homeomorphic to  $\mathbb{R}^{k-1} \times [0, \infty)$ ,
3. or, for manifolds with corners,  $\vec{p}$  has a neighborhood homeomorphic to  $\mathbb{R}^{k-l} \times [0, \infty)^l$ , with  $1 < l \leq n$ .

Points  $\vec{p} \in X$  that satisfy the first condition in the definition are called manifold points, and points that satisfy the second condition are called boundary points. Points that satisfy the third condition are called corner points. Usually,  $\bar{X}$  is compact, and the boundary is defined as  $\partial X = \bar{X} - \text{int } X$ . Note that  $\partial(\partial X) = \emptyset$ . A compact manifold  $X$  with  $\partial X = \emptyset$  is called a closed manifold. Note that a sphere is a closed manifold, a cylinder (without the circular caps) is not a closed manifold, and an infinite cylinder is not a closed manifold because it is not compact.

For two one-to-one maps of class  $C^1$ ,  $H_1$  from  $U_1 \in \mathbb{R}^k \rightarrow X$  and  $H_2$  from  $U_2 \in \mathbb{R}^k \rightarrow X$ , where  $X$  is a  $k$ -dimensional manifold, a reparametrization is a map  $T : \mathbb{R}^k \rightarrow \mathbb{R}^k$ , where  $T(U_1) = U_2$  so that  $H_2 \circ T = H_1$ . It can be proven that  $T$  is a  $C^1$  diffeomorphism.

Example in  $\mathbb{R}^3$  Consider a surface  $X$  in  $\mathbb{R}^3$ . Consider

$$H(u, v) = (x_1(u, v), x_2(u, v), x_3(u, v))$$

Then, denote  $\vec{T}_u = \left( \frac{\partial x_1}{\partial u}, \frac{\partial x_2}{\partial u}, \frac{\partial x_3}{\partial u} \right) \Big|_{(u_0, v_0)}$ , and  $\vec{T}_v = \left( \frac{\partial x_1}{\partial v}, \frac{\partial x_2}{\partial v}, \frac{\partial x_3}{\partial v} \right) \Big|_{(u_0, v_0)}$ , which are the columns of  $DH(u_0, v_0)$ . In  $\mathbb{R}^3$ , we can consider  $\vec{a} \times \vec{b} =$

$$\begin{vmatrix} \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} \in \mathbb{R}^3. \text{ This cross product satisfies the following properties.}$$

$\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$ ,  $\vec{a} \times \vec{b} \perp \vec{a}$ , and  $\vec{a} \times \vec{b} \perp \vec{b}$ . Then, we can define  $\vec{N} = \frac{\vec{T}_u \times \vec{T}_v}{\|\vec{T}_u \times \vec{T}_v\|}$ .

A continuous choice of  $\vec{N}$  at every point in  $X$  gives an orientation of  $X$ . Not all manifolds are orientable however. For instance, the mobius strip is non-orientable. Every (manifold) closed surface in  $\mathbb{R}^3$  is orientable. We wish to generalize the cross product to higher dimensions.

Wedge Product Let  $\mathbb{R}^n$  have standard basis  $\{\vec{e}_1, \dots, \vec{e}_n\}$ . Define  $\vec{e}_i \wedge \vec{e}_i = \vec{0}$ , and  $\vec{e}_i \wedge \vec{e}_j = -\vec{e}_j \wedge \vec{e}_i$ . For any  $I \subseteq \{1, \dots, n\}$ , where  $I = \{i_1, \dots, i_k\}$ , where  $|I| = k$ , where  $i_1 < i_2 < \dots < i_k$ , define  $\vec{e}_I = \vec{e}_{i_1} \wedge \vec{e}_{i_2} \wedge \dots \wedge \vec{e}_{i_k}$ . Define  $\wedge^k \mathbb{R}^n$  to be the vector space with basis  $\vec{e}_I$ , over all  $|I| = k$ . For  $|I| = k$  and  $|J| = k$ , we have

$$\vec{e}_I \wedge \vec{e}_J = \begin{cases} 0, & I \cap J \neq \emptyset \\ \vec{e}_{I \cup J}, & I \cap J = \emptyset \end{cases}$$

We have  $\dim(\wedge^k \mathbb{R}^n) = \binom{n}{k}$ . Denote  $\wedge^0 \mathbb{R}^n$  to be with basis  $\vec{e}_\emptyset$ , where  $\dim(\wedge^0 \mathbb{R}^n) = 1 \simeq \mathbb{R}$ . We also have  $\dim(\wedge^n \mathbb{R}^n) = 1 \simeq \mathbb{R}$ . We have  $\wedge^k \mathbb{R}^n \simeq \wedge^{n-k} \mathbb{R}^n$ . Note that  $\wedge^k \mathbb{R}^n = \vec{0}$  if  $k > n$ . Note that this generalizes the  $n = 3$  case of the cross product.

## 5 Lecture 4/16/24

Here we are going to have  $\mathbb{R}^n e_1, \dots, e_n$ . Now we define a new vector space  $\wedge^k \mathbb{R}^n, k = 0, \dots, n$  with  $\dim(\wedge^k \mathbb{R}^n) = \binom{n}{k}$  with basis  $e_{i_1} \wedge \dots \wedge e_{i_k}$ . Now from here

Ask Eric for his notes at the beginning of class.

If  $k \in \{0, \dots, n\}$   $\wedge^k \mathbb{R}^n \approx \wedge^{n-k} \mathbb{R}^n$  Here we wish to define an isomorphism that goes between these two spaces. The map Hodge map is a map that goes from  $\wedge^k \mathbb{R}^n \rightarrow \wedge^{n-k} \mathbb{R}^n$ . Here the basis vectors are going to be denoted as  $k = I, e_I \rightarrow \pm e_{I^c} = [1, \dots, n]/I$  The choice of the sign is denoted to be where  $e_I \wedge e_{I^c} = e_1 \wedge \dots \wedge e_n$  This map in  $\mathbb{R}^3$  is going to be defined by the map  $\wedge^0 \mathbb{R}^3 \rightarrow \wedge^3 \mathbb{R}^3$  and we see that the basis vectors are going to be the following between  $\wedge^2 \mathbb{R}^3 \rightarrow \wedge^1 \mathbb{R}^3$ .

Now for vectors  $v, w \in \mathbb{R}^3, *(v \wedge w) = v \times w$  Now thus we have the following idea that  $*(v_1 \wedge \dots \wedge v_{n-1}) = v_1 \times v_2 \times \dots \times v_{n-1}$ . Here we have generalized cross products to understand what they truly are. Here we have talked about two things: norm and orthogonality. However, we need to add more structure to what this is going to be, specifically a dot product or inner product.

**Inner Product** We will find a basis that makes the vectors orthonormal. Let us define the following inner product on  $\bigwedge^k \mathbb{R}^n$  to be a map that takes  $\bigwedge^k \mathbb{R}^n \times \bigwedge^k \mathbb{R}^n \rightarrow \mathbb{R} = \bigwedge^0 \mathbb{R}$ . For two vectors  $v, w$  we get that the norm is going to be  $\langle v, w \rangle = *(v \wedge *w)$ . In order to ensure that this is an inner product, we must evaluate the symmetries and the properties that exist for an inner product.

From here, let us verify that  $\langle e_i, e_j \rangle$  are orthonormal. From here we get that

$$*(e_I \wedge *e_I) = *(e_1 \wedge \dots \wedge e_n) = 1, I = J$$

$$*(e_I \wedge *e_J) = 0, I \neq J$$

Therefore this tells us this is an orthonormal basis. Therefore we can define a norm on  $\bigwedge^k \mathbb{R}^n$  where the norm squared  $|v|^2 = *(v \wedge *v)$  where  $v = \sum_I a_I e_I$  where  $|v| = \sqrt{\sum_I a_I^2}$

**Area** For  $v, w \in \mathbb{R}^n$  and  $P = tv + sw, t, s \geq 1$  and we wish to find the area between these two vectors. We will have that  $\text{Area}(P) = \|v \wedge w\|$  where this is in the space  $\bigwedge^2 \mathbb{R}^n$  and  $v \wedge w = \sum \det(2 \times 2 \text{ submatrix}) e_i \wedge e_j$

**Volume** We will have vectors  $v_1 \dots v_k \in \mathbb{R}^n$  and  $P = \sum_{u=1}^k t_u v_u, 0 \leq t_u \leq 1$  where  $P$  is a  $k$  dimensional parallelipiped in  $\mathbb{R}^n$  spanned by the vectors  $v_1 \dots v_k$ . The volume is thus going to be  $\text{Vol}(P) = \|v_1 \wedge \dots \wedge v_k\|$  with a norm on  $\bigwedge^k \mathbb{R}^n$

Suppose that we have a function  $r : [a, b] \rightarrow \mathbb{R}^n$ , smooth  $-C^1$  function at least. We want that  $v(t) \neq 0$ . If we let  $C$  be the image of the curve subset of  $\mathbb{R}^n$ , and we wish to measure the length of the curve, we get that

$$l(C) = \sum_{\text{partition}} \|r(t) - r(t - \delta t)\| = \int_a^b \|v(t)\| dt$$

## 6 Lecture 4/23/24

**Review of 4/19/24** We covered line integrals for smooth paths where they must be piece wise smooth for  $r : [a, b] \rightarrow \mathbb{R}^n$ . We also require that the speed is nonzero. The line integral has an initial point which we can denote

as  $\mathbf{r}(a)$  and an endpoint denoted as  $\mathbf{r}(b)$ . We define  $C$  to be the image of the path in  $\mathbb{R}^n$ . Now let us consider  $f : \mathbb{R}^n \rightarrow \mathbb{R}$  to be

$$\int f ds = \int_a^b f(\mathbf{r}(t)) \|\mathbf{r}'(t)\| dt$$

This is going to be a line integral for a scalar function over a curve. We are interested in vector spaces which we can denote as  $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$  which we can denote as

$$\int_C \mathbf{F} \cdot d\mathbf{s} = \int_C \mathbf{F} \cdot \mathbf{T} ds = \int_a^b \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t) dt$$

where  $T$  is the unit tangent vector.

**Definition**  $\mathbf{F}$  is a conservative vector field if  $\int \mathbf{F} \cdot d\mathbf{s}$  is path independent, i.e. only dependent on the beginning and end points.

Gradient Fields that go from  $n$  dimensional to one dimensional and have that  $\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}$ , we say that  $\mathbf{F}$  is a gradient field if  $\mathbf{F} = \nabla f$  of some  $f$  where  $f$  is denoted as a potential function for  $\mathbf{F}$

**Fundamental Theorem of Line Integrals** If  $C$  is given some parameterization of some line to  $\mathbb{R}^n$  which is smooth, then we have that FTLI states that

$$\int \nabla f \cdot d\mathbf{s} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

Now we define that if  $\mathbf{F} = \nabla f \implies \mathbf{F}$  is conservative. To evaluate the backwards direction, we have the following proposition.

For  $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ , continuous, then the following are equivalent

1.  $\mathbf{F}$  is a gradient field
2.  $\mathbf{F}$  is conservative
3.  $\int_C \mathbf{F} = 0$ , for any closed smooth path

To prove that (ii) and (iii), construct a contour  $C$ , if the contour  $C$  is conservative then we have that by the FTLI,  $\int_C \nabla f \cdot d\mathbf{s} = f(\mathbf{r}(a)) - f(\mathbf{r}(a)) = 0$

To show that (ii)  $\implies$  (i), over vector field is conservative and we wish to show it is the gradient of some small  $f$ . We can fix a point  $\mathbf{p}_o \in \mathbb{R}^n$ . We can define  $f(x) = \int_{\mathbf{p}_o}^x \mathbf{F} \cdot d\mathbf{s} = \int_C \mathbf{F} \cdot d\mathbf{s}$  where  $C$  connects  $\mathbf{p}_o$  to  $x$ . Let us choose a different path denoted as  $C$ , we can see that this path integrated will result in the same value of the integral of  $C$ , which shows that the integral over  $\mathbf{F}$  is well defined.

We will consider the following argument now. For a path we can denote as  $C_1$ , the path will be directly under our endpoint with distance  $h$  away given by a ball defined to be  $h$  large. Thus we see that to get to our endpoint, we will have that  $x + \epsilon e_i$  where  $e_i$  is our basis vector. Therefore we now know that for the directional derivative, it is defined as

$$\frac{\partial f}{\partial x_i} = \lim_{h \rightarrow 0} \frac{f(x + h e_i) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{1}{h} \left( \int_{\mathbf{p}_o}^{x + h e_i} \mathbf{F} - \int_{\mathbf{p}_o}^x \mathbf{F} \right)$$

Consider  $r(t) = x + t e_i, t \in [0, h]$ . Thus we can look at the integral

$$\lim_{h \rightarrow 0} \frac{1}{h} \int_0^h \mathbf{F}(r(t)) \mathbf{r}'(t) dt = \lim_{h \rightarrow 0} \frac{1}{h} \int_0^h \mathbf{F}(r(t)) e_i dt$$

Therefore by the fundamental theorem of calculus, we have that this is going to be equal to  $\mathbf{F}_i(x)$

**Remark** The previous proposition holds if  $\mathbf{F} : U \rightarrow \mathbb{R}^n$  where  $U$  is an open and convex subset of  $\mathbb{R}^n$

**Theorem**  $\mathbf{F} : \mathbb{R}^n \rightarrow \mathbb{R}^n$  is of class  $C^1$ , then  $\mathbf{f}$  is conservative iff  $\frac{\partial F_i}{\partial x_j} = \frac{\partial F_j}{\partial x_i}$  hold for all  $i, j$

**Proof** For the forward direction, we know that  $\mathbf{F} = \nabla f$  therefore through Clairot's, we can show that these are equal by taking the second derivative of the terms.

For the backwards direction, because  $0 \in \mathbb{R}^n, \forall x \in \mathbb{R}^n, t x \in \mathbb{R}^n, t \in [0, 1]$  then we have that

$$f(x) = \sum_{j=1}^n x_j \int_0^1 \mathbf{F}_j(t \mathbf{x}) dt$$

Thus we wish to show that  $\mathbf{F} = \nabla f, \frac{\partial f}{\partial x_i} = F_i$

## 6.1 Surface Integrals

$\Sigma \subset \mathbb{R}^3$  smooth surface with parameterization  $H : D \rightarrow \mathbb{R}^3, D \subset \mathbb{R}^2$  which would be a linear mapping/operator that takes us from 2 dimensions to three dimensions. Now we wish to ask what  $\arg Area(\Sigma) \int \int_{\Sigma} dA, dA = ||T_u \times T_v|| du dv$ . We can visualize it simply as a transformation of our  $\mathbb{R}^2$  space with unit vectors  $u$  and  $v$  that undergo a linear transformation denoted as  $T_u$  and  $T_v$  which can be thought of the area s.t.  $T_v \Delta v$  and  $T_u \Delta u$  are going to be our approximated lengths in the  $\mathbb{R}^3$  space.

Now if there exists some space in higher dimensions, we can write this area as  $||T_u \wedge T_v||$

Next time, we will understand how to integrate a scalar field, and what it means to integrate a vector field.

## 7 Lecture 4/25/24

Let's have a smooth surface  $\Sigma \subset \mathbb{R}^3$  with a parameterization. Thus we will get a tangent vector that we can denote as  $T_v$  and  $T_u$  that will allow us to transfer from  $(u, v) \rightarrow H(u, v) = (x(u, v), y(u, v), z(u, v))$ . Thus we will have that

$$T_v = \left( \frac{\partial x}{\partial v}, \frac{\partial y}{\partial v}, \frac{\partial z}{\partial v} \right), T_u = \left( \frac{\partial x}{\partial u}, \frac{\partial y}{\partial u}, \frac{\partial z}{\partial u} \right)$$

Thus we will have that

$$\Delta S = \Delta A = ||T_u \times T_v|| \Delta u \Delta v, \lim_{u,v} \rightarrow 0$$

which  $\Delta S$  denotes the scalar surface area element which should be equal to the scalar flat surface area.

$$T_u \times T_v = (\partial(y, z)/\partial(u, v), \partial(x, z)/\partial(u, v), \partial(x, y)/\partial(u, v))$$

which is offhand notation for the cross product elements.

Now recall from last time (Fubini's Theorem) that

$$Area(\Sigma) = \int \int_{\Sigma} dS = \int \int_D ||T_u \times T_v|| du dv$$

$f : \mathbb{R}^3 \rightarrow \mathbb{R}$  continuous, we define the integral  $\int \int_{\Sigma} f dS$  which we can think of this as a mass integral

## 7.1 Flux Integrals

We can consider the following integral of flux to be

$$\int \int_{\Sigma} \mathbf{F} \cdot \mathbf{n} dS, \mathbf{n} = \frac{T_u \times T_v}{||T_u \times T_v||}$$

where  $\mathbf{F}$  is our vector field. Thus we will get after transformation that

$$\int \int_D F(x(u, v), y(u, v), z(u, v)) \cdot (T_u \times T_v) du dv$$

Remark: Flux integrals are independent of parameterization up to orientation

**Jordan Separation Theorem**  $C$  is a simple closed curve such that

$$\mathbb{R}^2/C = U_1 \cup U_2$$

$U_1$  and  $U_2$  open, connected such that  $U_1$  is bounded and  $U_2$  is unbounded.

For simple closed surfaces,  $\Sigma$ , we can see that a torus also obeys the Jordan separation theorem. We can see that for the unit normal vector  $\mathbf{n}$ , the vector that points outward oriented is by convention going to be the positive orientation.

## 8 Final Lecture 4/30/24

Recall the Fundamental Theorem of Calculus for integrals. We can generalize this to FTLI or the fundamental theorem of line integrals in which we can understand it as

$$\int_C \nabla f \cdot d\mathbf{r} = f(\mathbf{r}(b)) - f(\mathbf{r}(a))$$

When we generalize this into 2 dimensions, we will have Green's Theorem. For example, consider a positively oriented (counterclockwise) closed curve  $C$ . This separates by Jordan Separation Theorem into an inside and outside. We understand Green's Theorem that



$$\int \int_D (\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}) dx dy = \int_{surface} P dx + Q dy$$